



Project 15 Test-Retest Reliability of Semi-quantitative Readings from Knee Radiographs

1. SAS dataset	1
2. Purpose and methods	1
2.1 Background	1
2.2 Reliability at single time points	1
2.3 Longitudinal reliability	2
3. Results	2
4. References	3
5. Tables	4

1. SAS dataset

Name: **kXR_SQ_Rel_BUxx** (xx identifies the time point)

Display label: **kXR SQ reliability readings (BU)**

2. Purpose and methods

2.1 Background

Project 15 (ARRA Radiologic Outcomes for the OAI) consists of semi-quantitative readings of knee radiographs done at the Boston University under the direction of Dr. David Felson. To evaluate test-retest reliability the complete set of films (all time points) of 150 participants were sent by the Coordinating Center to the reading center after the original readings were completed. The reading center was blinded to the fact that these were retest readings and blinded to the original scores of the readings. The sample was representative of the entire cohort with respect to the grades of OA, radiographic progression and incidence and number of time points with radiographs for a participant. The reading protocols for the first and second readings were the same, including the initial screening, or triage, reading, identification of discrepancies and adjudication procedures. The first and second sets of measurements were separated by 3 to 9 months.

Both original and retest readings are included in the kXR_SQ_Rel_BUxx datasets, and are denoted by their project number in the READPRJ variable. Records with READPRJ=19A are the original readings (which can also be found in the kXR_SQ_BUxx datasets that contain all the original Project 15 ARRA data), and records with READPRJ=19B are the retest readings. Project 19B data replace the reliability data previously released (Project 14).

2.2 Reliability at single time points

To test reliability at each time point we evaluated each reading variable by creating a cross tabulation of the variable from the two readings (2 knees per participant). Simple kappa coefficients were used to evaluate agreement between the two readings when the variable was dichotomous. Weighted kappa coefficients were used if the variable had more than two ordinal categories. The kappa coefficient is a measure of agreement (Sim and Wright 2005); a value >0 indicates that the agreement observed is greater than that expected by chance and a value of 1.0 indicates perfect agreement. We do not include results for several OA features that did not have sufficient observations with values scored >0 to yield valid results.

We also created several calculated dichotomous variables for the presence/absence of a feature and the severity of a feature and evaluated their agreement as described above. The calculated variables are included in the reliability analysis but are not included in the released dataset.

For all single time point joint space narrowing (JSN) variables, if the JSN score was a non-integer value (e.g. x.3, x.6, x.9) we rounded that value down to the closest integer.



2.3 Longitudinal reliability

For longitudinal reliability we performed the following analyses.

We compared the results of the initial screening (or triage) reading which determined whether the longitudinal readings included just the K-L grades and JSN (subjects who were screened negative for having K-L grade ≥ 2 in either knee at any time point) or included K-L grades, JSN and all other individual radiographic features (subjects who were screened positive for having K-L grade ≥ 2 in at least one knee at one or more time points).

For the two different definitions of incident radiographic knee OA (see section 2.4a in the kXR_SQ_BU_xx documentation) we first compared the two readings for whether a knee was eligible for incidence, which for both definitions required that the knee have K-L grade of 0 or 1 at baseline. For incident OA using each definitions at a given follow-up time point, among those with reading data at that follow-up visit we first compared the overall classification of the knee (eligible for incident OA at baseline but no incident OA during follow-up; incident OA during follow-up; prevalent OA at baseline and not eligible for incident OA). Second, restricting the comparison to the knees that were classified as eligible for incident radiographic OA by both readings, we compared the classification for incident OA up to that time point between readings. We also performed similar comparisons between readings for variables summarizing incident knee OA status across all time points in which we used the reading data from the last time point available and all earlier time points to classify a knee for incident OA.

For progression of JSN, we analyzed medial and lateral JSN separately. Analyses included comparison between readings for whether there was any increase (progression) in JSN scores up to a given follow-up time point, counting within grade increases in JSN as progression, among knees with reading data at that time point. We also summarized JSN increases over all available time points in which we used the reading data from the last time point available and all earlier time points to classify a knee for JSN progression.

3. Results

Cross-sectional reliability

The kappa coefficients for OA features assessed on the baseline and the 36 months radiographs are presented in Table 1 below. Tables 2 to 4 have examples of the cross tabulations between the reading scores for selected variables.

The Kellgren-Lawrence grade and JSN variables had good agreement between the two readings, with kappa coefficient values of approximately 0.70 to 0.80 and 0.75 to 0.88, respectively. The osteophyte variables showed good, but slightly lower, agreement, with kappas ranging from 0.69 to 0.82. The sclerosis variables also had good but lower agreement, with kappas of 0.65 to 0.85 for the raw variables and 0.73 to 0.79 for a summary variable for the presence/absence sclerosis over all four regions scored. Agreement was good to excellent for presence/absence of chondrocalcinosis in each compartment, with kappas of 0.79 to 0.91. Agreement on whether any subchondral cysts were present was moderate, with kappas of 0.45 to 0.62. Agreement on test-retest readings of 12-, 24- and 48-month radiographs were very similar to that for baseline and 36 months.

Longitudinal reliability

Longitudinal reliability results for incident radiographic knee OA and JSN progression are detailed in tables 5-21 in section 4. Few longitudinal studies have provided data on the reliability of either incidence or JSN progression, so these OAI data are somewhat unique.

There was good to excellent agreement on baseline radiographic knee OA status classifications that determined which longitudinal features were scored and whether a knee was eligible to be classified as developing incident OA during follow-up. Good to excellent agreement was also found for both definitions of incident OA among knees that were classified as eligible for incidence in both readings. Reliability for incidence tended to improve with increased duration of follow-up. However, a greater number of knees were classified as eligible for incident OA on the first



reading compared to the second reading, suggesting that there was some shift in the application of the threshold for KL grade 2 and KL grade 2N that would affect the classification for incident OA.

Reliability for JSN progression was good to excellent for both medial and lateral compartments. As with incidence, reliability tended to increase with longer follow-up times.

4. References

1. Sim, J and Wright, C. The kappa statistic in reliability studies: use, interpretation, and sample size requirements. *Physical Therapy*. Volume 85. Number 3. March 2005.



5. Tables

Table 1: Agreement between first and second readings (Project 15)			
Variable Name	Variable Label	Kappa* (95%CI)	Kappa* (95%CI)
		Baseline	36 month
VxxXRKL	BL/FU kXR reading (BU): Kellgren and Lawrence (grades 0-4)	0.70 (0.65-0.76)	0.78 (0.73-0.83)
KL_OA	KL_OA: VxxXRKL <2=0; VxxXRKL ≥2=1	0.70 (0.62-0.78)	0.80 (0.72-0.87)
VxxXRJSL	BL/FU kXR reading (BU): joint space narrowing (OARSI grades 0-3) lateral	0.87 (0.76-0.98)	0.85 (0.76-0.94)
ANY JSNL	ANY JSNL: VxxXRJSL <1=0; VxxXRJSL ≥1=1	0.83 (0.70-0.96)	0.88 (0.77-0.98)
VxxXRJSM	BL/FU kXR reading (BU): joint space narrowing (OARSI grades 0-3) medial	0.75 (0.68-0.81)	0.77 (0.71-0.83)
ANY JSNM	ANY JSNM: VxxXRJSM <1=0; VxxXRJSM ≥1=1	0.74 (0.66-0.82)	0.75 (0.67-0.83)
VxxXROSFL	BL/FU kXR reading (BU): osteophytes (OARSI grades 0-3) femur lateral	0.69 (0.60-0.79)	0.73 (0.65-0.82)
VxxXROSFM	BL/FU kXR reading (BU): osteophytes (OARSI grades 0-3) femur medial compartment	0.73 (0.66-0.81)	0.82 (0.76-0.82)
ANY FEMOST	ANY FEMOST: VxxXROSFL <1 and VxxXROSFM <1=0; VxxXROSFL ≥1 or VxxXROSFM ≥1=1	0.78 (0.70-0.85)	0.81 (0.74-0.88)
VxxXROSTL	BL/FU kXR reading (BU): osteophytes (OARSI grades 0-3) tibia lateral compartment	0.70 (0.61-0.79)	0.71 (0.62-0.80)
VxxXROSTM	BL/FU kXR reading (BU): osteophytes (OARSI grades 0-3) tibia medial compartment	0.69 (0.60-0.77)	0.73 (0.66-0.81)
ANY TIBOST	ANY TIBOST: VxxXROSTL <1 and VxxXROSTM <1=0; VxxXROSTL ≥1 or VxxXROSTM ≥1=1	0.71 (0.63-0.79)	0.74 (0.65-0.82)
VxxXRSCFL	BL kXR reading (BU): sclerosis (OARSI grades 0-3) tibia lateral compartment	0.67 (0.50-0.85)	0.79 (0.68-0.90)
VxxXRSCFM	BL/FU kXR reading (BU): sclerosis (OARSI grades 0-3) femur medial compartment	0.68 (0.59-0.78)	0.79 (0.71-0.76)
VxxXRCTL	BL kXR reading (BU): sclerosis (OARSI grades 0-3) tibia lateral compartment	0.65 (0.49-0.82)	0.85 (0.73-0.96)
VxxXRCTM	BL kXR reading (BU): sclerosis (OARSI grades 0-3) tibia medial compartment	--	0.76 (0.68-0.84)



Table 1: Agreement between first and second readings (Project 15)

Variable Name	Variable Label	Kappa* (95%CI)	Kappa* (95%CI)
ANY SCLER	ANY SCLER: VxxXRSCFL<1 and VxxXRSCFM<1 and VxxXRCTL<1 and VxxXRSCFM<1 = 0; VxxXRSCFL ≥1 or VxxXRSCFM ≥1 or VxxXRCTL≥1 or VxxXRSCFM≥1 = 1	0.73 (0.64-0.82)	0.79 (0.78-0.99)
ANY CYST	ANY CYST: VxxXRCYFL<1 and VxxXRCYFM<1 and VxxXRCYTL<1 and VxxXRSCFM<1 = 0; VxxXRSCFL ≥1 or VxxXRSCFM ≥1 or VxxXRCTL≥1 or VxxXRSCFM≥1 = 1	0.62 (0.40-0.85)	0.45 (0.24-0.66)
VxxXRCHL	BL/FU kXR reading (BU): chondrocalcinosis (0-1) femur lateral compartment	0.85 (0.68-1.00)	0.89 (0.75-1.00)
VxxXRCHM	BL/FU kXR reading (BU): chondrocalcinosis (0-1) femur lateral compartment	0.79 (0.59-0.99)	0.91 (0.79-1.00)



Indicates calculate variables used in this report that are not included in a released dataset

* Simple kappa statistic for dichotomous variables, weighted kappas for variables with three or more ordinal categories



Table 2: Test-retest agreement between Kellgren and Lawrence grade (VxxXRKL) at the baseline visit and 36 month visit for Project 15.

	Project 15 – baseline KLG Score 2						
Project 15 Baseline KLG Score		0	1	2	3	4	Total
	0	73	22	5	0	0	100
	1	20	28	25	0	0	73
	2	2	11	62	5	0	80
	3	0	1	8	36	1	46
	4	0	0	0	0	1	1
	Total	95	62	100	41	2	300

Kappa Coefficient = 0.70 (95%CI: 0.65-0.76)

	Project 15 – 36 Month KLG Score 2						
Project 15 36 Month KLG Score		0	1	2	3	4	Total
	0	53	10	3	0	0	66
	1	9	18	7	1	0	35
	2	3	10	75	6	0	94
	3	0	1	13	39	0	53
	4	0	0	0	3	9	12
	Total	65	39	98	49	9	260

Kappa Coefficient = 0.78 (95%CI: 0.73-0.83)



Table 3: Test-retest agreement between joint space narrowing lateral compartment (VxxXRJSL) at the baseline visit and 36 month visit for Project 15.

Project 15 Baseline JSN -L						
Project 15 Baseline JSN -L		0	1	2	3	Total
	0	278	3	0	0	281
	1	2	7	0	0	9
	2	1	0	9	0	10
	3	0	0	0	0	0
	Total	281	10	9	0	300

Kappa Coefficient = 0.87 (95%CI: 0.76-0.98)

Project 15 36 Month JSN -L						
Project 15 36 Month JSN -L		0	1	2	3	Total
	0	235	2	0	0	237
	1	2	2	2	0	6
	2	1	3	9	0	13
	3	0	0	1	3	4
	Total	238	7	12	3	260

Kappa Coefficient = 0.85 (95%CI: 0.76-0.94)



Table 4: Test-retest agreement between joint space narrowing medial compartment (VxxXRJSM) at the baseline visit and 36 month visit for Project 15.

		Project 15 Baseline JSN -M				
Project 15 Baseline JSN -M		0	1	2	3	Total
	0	170	10	0	0	180
	1	27	50	5	0	82
	2	0	9	27	1	37
	3	0	0	0	1	1
	Total	197	69	32	2	300

Kappa Coefficient = 0.75 (95%CI: 0.68-0.81)

		Project 15 36 month JSN -M				
Project 15 36 Month JSN -M		0	1	2	3	Total
	0	141	8	0	0	149
	1	22	35	5	0	62
	2	1	10	30	0	41
	3	0	0	2	6	8
	Total	164	53	37	6	260

Kappa Coefficient = 0.77 (95%CI: 0.71-0.84)



Table 5.

Variable	Kappa* (95%CI)	Kappa* (95%CI)	Kappa* (95%CI)	Kappa* (95%CI)	Kappa* (95%CI)
	Baseline				
Screening reading for IRF scoring (0=KL 0-1 both knees all time points; 1=KL ≥2 either knee at any time point)	0.79 (0.66–0.91)	NA	NA	NA	NA
Knee is eligible for incident OA at baseline: 0=KL ≥2; 1=KL 0-1)	0.71 (0.62–0.79)	NA	NA	NA	NA
	12 month	24 month	36 month	48 month	At any** time point
Incident KL ≥2 OA in knees score K-L 0-1 at baseline, both readings (0=No; 1=Yes)	0.64 (0.38–0.91)	0.60 (0.37–0.84)	0.76 (0.61–0.91)	0.82 (0.69–0.96)	0.71 (0.55–0.87)
Incident KL ≥2N OA in knees score K-L 0-1 at baseline, both readings (0=No; 1=Yes)	0.70 (0.60–0.95)	0.80 (0.66–0.94)	0.76 (0.61–0.91)	0.82 (0.70–0.94)	0.81 (0.70–0.92)
Medial JSN progression: ≥ within grade increase from baseline in knees with follow-up at each time point (0=No; 1=Yes)	0.64 (0.48–0.79)	0.67 (0.54–0.79)	0.75 (0.63–0.86)	0.78 (0.68–0.88)	0.72 (0.63–0.82)
Lateral JSN progression: ≥ within grade increase from baseline in knees with follow-up at each time point (0=No; 1=Yes)	0.39 (0.1–0.78)	0.73 (0.54–0.92)	0.74 (0.56–0.91)	0.85 (0.70–0.99)	0.83 (0.70–0.96)
Medial JSN progression severity (0=None; 1=within-grade only; 2= full grade or more)***	NA	NA	NA	NA	0.71 (0.62–0.80)
Lateral JSN progression severity (0=None; 1=within-grade only; 2= full grade or more)***	NA	NA	NA	NA	0.79 (0.64–0.93)

Calculated variables used in this table are not included in a released dataset.

* Simple kappa statistic for dichotomous variables, weighted kappas for variables with three or more ordinal categories.

** Cumulative changes, with last available time point carried forward.

***For 3 level progression severity variable, the greatest increase at any time point is used to classify the knee.



Table 6. Screening reading: K-L grade ≥ 2 in either knee at any time point (Yes/No)
 (Subjects screened negative were read for K-L grade and JSN scores in both knees at all time points.
 Subjects screened positive had K-L grade, JSN and all other IRFs at each time point.)

K-L grade $\geq$ at any time point			
Reading 1	Reading 2		Total
Frequency	N	Y	
N	24	3	27
Y	7	116	123
Total	31	119	150

Kappa Coefficient = 0.79 (95% CI: 0.66 – 0.91)

Table 7. Knee is eligible at baseline for incident OA during follow-up: K-L grade 0-1 (eligible) vs K-L grade ≥ 2 (not eligible)

K-L grade at baseline			
Reading 1	Reading 2		Total
Frequency	0-1	2-4	
0-1	143	30	173
2-4	14	113	127
Total	157	143	300

Kappa Coefficient = 0.71 (95% CI: 0.62 – 0.79)



Table 8. Incident knee OA by 24 months, *K-L grade* ≥ 2 definition: knee is KL 0-1 at baseline and K-L grade ≥ 2 by 24 months or an earlier time point

Incident knee OA by 24 mo: K-L ≥ 2				
Reading 1	Reading 2			Total
Frequency	K-L 0-1 at baseline No change	K-L 0-1 at baseline Incident OA	K-L ≥ 2 at baseline	
K-L 0-1 at baseline No change	109	1	19	129
K-L 0-1 at baseline Incident OA	8	8	9	25
K-L ≥ 2 at baseline	10	2	110	122
Total	127	11	138	276

Overall Kappa Coefficient = 0.68 (95% CI: 0.61– 0. 76)

Overall Weighted Kappa Coefficient = 0.72

Kappa Coefficient for knees eligible for incident OA by both readings (shaded cells = 0.60 (95% CI: 0.37, 0.84)

Table 9. Incident OA by 24 months, *K-L grade* $\geq 2N$ definition: knee is KL 0-1 at baseline and K-L grade $\geq 2N$ by 24 months or an earlier time point, in knees with reading data at 48 months.

Incident knee OA by 24 mo: K-L $\geq 2N$				
Reading 1	Reading 2			Total
Frequency	K-L 0-1 at baseline No change	K-L 0-1 at baseline Incident OA	K-L ≥ 2 at baseline	
K-L 0-1 at baseline No change	99	4	13	116
K-L 0-1 at baseline Incident OA	5	18	15	38
K-L ≥ 2 at baseline	9	3	110	122
Total	113	25	138	276

Overall Kappa Coefficient = 0.70 (95%CI: 0.63 – 0.78)

Overall Weighted Kappa Coefficient = 0.74

Kappa Coefficient for knees eligible for incident OA by both readings (shaded cells) = 0.80 (95% CI: 0.66, 0.94)



Table 10. Incident OA by 48 months, *K-L grade* ≥ 2 definition: knee is KL 0-1 at baseline and K-L grade ≥ 2 by 48 months or an earlier time point, in knees with reading data at 48 months.

Incident knee OA by 48m: K-L ≥ 2				
Reading 1	Reading 2			Total
Frequency	K-L 0-1 at baseline No change	K-L 0-1 at baseline Incident OA	K-L ≥ 2 at baseline	
K-L 0-1 at baseline No change	84	1	16	101
K-L 0-1 at baseline Incident OA	5	12	7	24
K-L ≥ 2 at baseline	8	3	96	107
Total	97	16	119	232

Overall Kappa Coefficient = 0.70 (95% CI: 0.62 – 0.78)

Overall Weighted Kappa Coefficient = 0.72

Kappa for knees eligible for incident OA by both readings (shaded cells) = 0.82 (95% CI: 0.69, 0.96)

Table 11. Incident OA by 48 months, *K-L grade* $\geq 2N$ definition: knee is KL 0-1 at baseline and K-L grade $\geq 2N$ by 48 months or earlier time point, in knees with reading data at 48 months.

Incident knee OA by 48 mo: K-L $\geq 2N$				
Reading 1	Reading 2			Total
Frequency	K-L 0-1 at baseline No change	K-L 0-1 at baseline Incident OA	K-L ≥ 2 at baseline	
K-L 0-1 at baseline No change	66	1	7	74
K-L 0-1 at baseline Incident OA	7	28	16	51
K-L ≥ 2 at baseline	7	4	96	107
Total	80	33	119	232

Overall kappa Coefficient = 0.71 (95% CI: 0.63 – 0.79)

Overall weighted kappa Coefficient = 0.75

Kappa Coefficient for knees eligible for incident OA by both readings (shaded cells) = 0.82 (95% CI: 0.70, 0.94)



Table 12. Incident OA by 48 months, *K-L grade ≥ 2* definition: knee is KL 0-1 at baseline and K-L grade ≥ 2 by the last time point with reading data or an earlier time point

Incident knee OA at any time point: K-L ≥ 2				
Reading 1	Reading 2			Total
Frequency	K-L 0-1 at baseline No change	K-L 0-1 at baseline Incident OA	K-L ≥ 2 at baseline	
K-L 0-1 at baseline No change	112	1	16	129
K-L 0-1 at baseline Incident OA	10	17	14	41
K-L ≥ 2 at baseline	11	4	113	128
Total	133	22	143	298

Overall kappa Coefficient = 0.68 (95% CI: 0.61 – 0.75)

Overall weighted kappa Coefficient = 0.72

Kappa Coefficient for knees eligible for incident OA by both readings (shaded cells) = 0.71 (95%CI: 0.55, 0.87)

Table 13. Incident OA by 24 months, *K-L grade $\geq 2N$* definition: knee is KL 0-1 at baseline and K-L grade $\geq 2N$ by the last time point with reading data or an earlier time point

Incident knee OA at any time point: K-L $\geq 2N$				
Reading 1	Reading 2			Total
Frequency	K-L 0-1 at baseline No change	K-L 0-1 at baseline Incident OA	K-L ≥ 2 at baseline	
K-L 0-1 at baseline No change	94	1	6	101
K-L 0-1 at baseline Incident OA	10	35	24	69
K-L ≥ 2 at baseline	9	6	113	128
Total	113	42	143	298

Overall kappa Coefficient = 0.70 (95% CI: 0.64 – 0.77)

Overall weighted kappa Coefficient = 0.75

Kappa Coefficient for knees eligible for incident OA by both readings (shaded cells) = 0.81 (95% CI: 0.70, 0.92)



Table 14. Medial JSN progression by 24 months: Any increase in JSN grade (within grade or $\geq$ full grade) from baseline to 24 months in knees with reading data at 24 months

Medial JSN increase by 24 months			
Reading 1	Reading 2		Total
Frequency	No change	JSN increase	
No change	226	9	235
JSN increase	14	29	43
Total	240	38	278

Kappa Coefficient = 0.67 (95% CI: 0.54 – 0.79)

Table 15. Lateral JSN progression by 24 months: Any increase in JSN grade (within grade or $\geq$ full grade) from baseline to 24 months in knees with reading data at 24 months

Lateral JSN increase by 24 months			
Reading 1	Reading 2		Total
Frequency	No change	JSN increase	
No change	261	2	263
JSN increase	5	10	15
Total	266	12	278

Kappa Coefficient = 0.73 (95% CI: 0.54 – 0.92)



Table 16. Medial JSN progression by 48 months: Any increase in JSN grade (within grade or $\geq$ full grade) from baseline to 48 months in knees with reading data at 48 months

Medial JSN increase by 48 months			
Reading 1	Reading 2		Total
Frequency	No change	JSN increase	
No change	173	6	179
JSN increase	12	43	55
Total	185	49	234

Kappa Coefficient = 0.78 (95% CI: 0.68 – 0.87)

Table 17. Lateral JSN progression by 48 months: Any increase in JSN grade (within grade or $\geq$ full grade) from baseline to 48 months in knees with reading data at 48 months

Lateral JSN increase by 48 months			
Reading 1	Reading 2		Total
Frequency	No change	JSN increase	
No change	218	2	220
JSN increase	2	12	14
Total	220	14	234

Kappa Coefficient = 0.85 (95% CI: 0.70 – 0.99)



Table 18. Medial JSN progression at any time point: Any increase in JSN grade (within grade or $\geq$ full grade) from baseline to the last time point with reading data or an earlier time point

Medial JSN increase at any time point			
Reading 1	Reading 2		Total
Frequency	No change	JSN increase	
No change	218	10	228
JSN increase	19	53	72
Total	237	63	300

Kappa Coefficient = 0.72 (95% CI: 0.63 – 0.82)

Table 19. Lateral JSN progression at any time point: Any increase in JSN grade (within grade or $\geq$ full grade) from baseline to the last time point with reading data or an earlier time point

Lateral JSN increase at any time point			
Reading 1	Reading 2		Total
Frequency	No change	JSN increase	
No change	278	3	281
JSN increase	3	16	19
Total	281	19	300

Kappa Coefficient = 0.83 (95% CI: 0.70 – 0.96)



Table 20. Medial JSN progression (no change, within grade or $\geq$ full grade) at any time point: Greatest increase in JSN grade (within grade or $\geq$ full grade) from baseline to the last time point with reading data or an earlier time point

Medial JSN increase at any time point				
Reading 1	Reading 2			Total
Frequency	No change	Within-grade increase	$\geq$ full grade increase	
No change	218	4	6	228
Within grade increase	7	5	4	16
$\geq$ full grade increase	12	5	39	56
Total	237	14	49	300

Weighted kappa Coefficient = 0.71 (95% CI: 0.62 – 0.80)

Table 21. Lateral JSN progression (no change, within grade or $\geq$ full grade) at any time point: Greatest increase in JSN grade (within grade or $\geq$ full grade) from baseline to the last time point with reading data or an earlier time point

Medial JSN increase at any time point				
Reading 1	Reading 2			Total
Frequency	No change	Within-grade increase	$\geq$ full grade increase	
No change	278	0	3	281
Within grade increase	1	4	0	5
$\geq$ full grade increase	2	2	10	14
Total	281	6	13	300

Weighted Kappa Coefficient = 0.79 (95% CI: 0.64 – 0.93)